

E.E. 3334: S01

Software Development (Tech-SAL)

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Project Description: Software Development

- To aid in the development in the different elements of the Tech-SAL Programming Language. Specifically:
 - Visual Tech-SAL GUI
 - The Compiler / Translator
 - Tech-SAL IDE
 - Language Capabilities
 - Hardware Control
 - Ease of Use
- Assist with the West Texas Best Program
 - Section 2 members
 - Electronics Scoring Software
 - Section 1 members
 - Return Kit

Project Overview

What is Tech-SAL?

- Texas Tech Semi-Autonomous Language
- Currently Version .7
- The Tech-SAL language is to be used with the MSP430 microprocessor and is used to program motors, servos, and sensors, without having knowledge of programming or knowledge of the MSP430.

-Taken from the Tech-SAL Readme

File
Sites

Tech-SAL

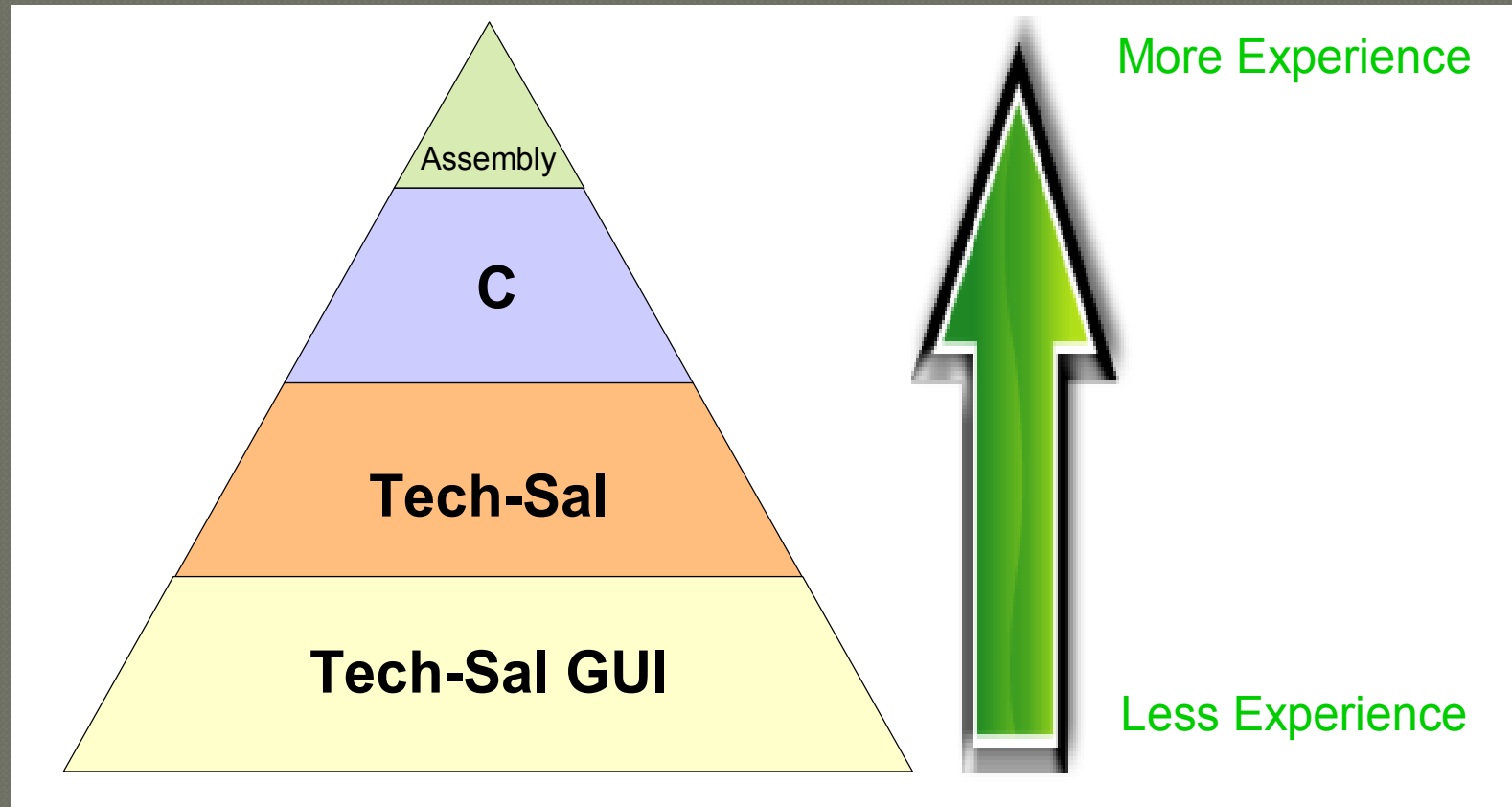
● Purpose of Tech-SAL:

- “To allow programming of MSP430 based microcontroller systems by presenting an interface that can be easily used with little programming knowledge or experience.”

● Tech-SAL consists of:

- Tech-SAL language specifications (AKA code rules)
- Tech-SAL translator
- Tech-SAL IDE

Tech-SAL



Tech-SAL: Start of Semester

- ◉ All Files Contained in .Zip Archive
- ◉ Features:
 - Tech-SAL Translator
 - Syntax Checker (Basic)
 - Visual Tech-SAL GUI
 - Readme File

Tech-SAL: End of Semester

- ◉ Redistributable .msi package

- Organized File Structure

- ◉ Features:

- Tech-SAL IDE
 - Tech-SAL Translator
 - Visual Tech-SAL GUI
 - Syntax Highlighting / Checking
 - .tsal File Extension
 - Full Documentation
 - Start Menu Directory

Installation Package

Tech-SAL '.msi' Package

- ◉ Built Using Visual Studio 2005
- ◉ Works with Microsoft Windows
- ◉ Installs to organized directory
- ◉ Handles Registry
 - Tech-SAL IDE
 - .tsal File Extension

Tech-SAL '.msi' Package



Tech-SAL IDE

Tech-SAL IDE: About

- ◉ Written in C# with Visual Studio 2005
- ◉ Designed to be Central Hub of Tech-SAL Project
- ◉ Incorporates all other components of project into one easy to use system.

Tech-SAL IDE: Features

- ◉ Open / Save .tsal Code Files
- ◉ Tech-SAL Syntax Highlighting
- ◉ Built in Translator Output
- ◉ Easy to Use Toolbar
- ◉ Tray Icon
- ◉ Access to all Documentation
- ◉ Visual Tech-SAL GUI Link
- ◉ SALConfig.txt File Control

Tech-SAL IDE: Features (Unused)

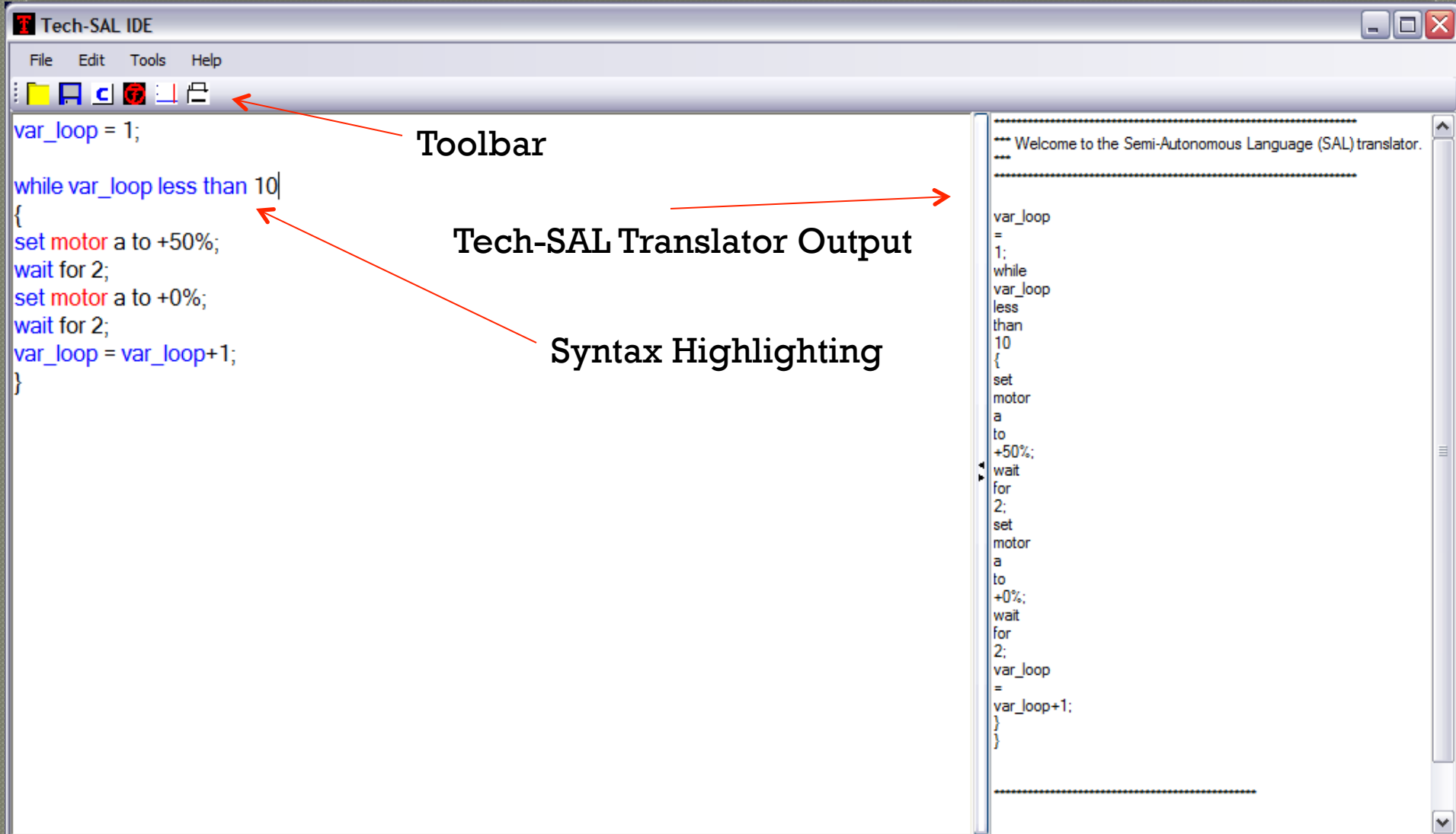
- ◉ Update Tool

- Provided in package with documentation, ready to be used

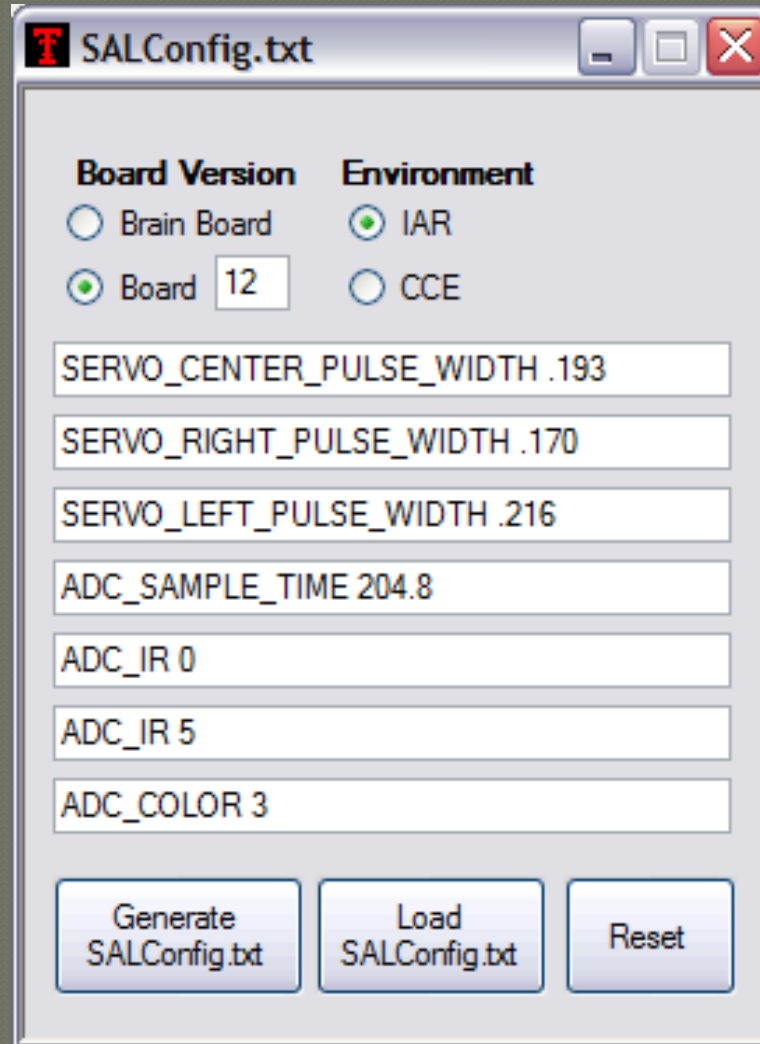
- ◉ Graphical Boot Strap Loader

- Coded but not fully tested or debugged

Tech-SAL IDE In Action



SALConfig.txt Menu



The screenshot shows a window titled "SALConfig.txt" with a standard Windows-style title bar (minimize, maximize, close buttons). The window contains two columns of radio button options: "Board Version" and "Environment". Under "Board Version", there are two options: "Brain Board" (unselected) and "Board 12" (selected). Under "Environment", there are two options: "IAR" (selected) and "CCE" (unselected). Below these options are seven text input fields containing the following values: "SERVO_CENTER_PULSE_WIDTH .193", "SERVO_RIGHT_PULSE_WIDTH .170", "SERVO_LEFT_PULSE_WIDTH .216", "ADC_SAMPLE_TIME 204.8", "ADC_IR 0", "ADC_IR 5", and "ADC_COLOR 3". At the bottom of the window are three buttons: "Generate SALConfig.txt", "Load SALConfig.txt", and "Reset".

SALConfig.txt

Board Version **Environment**

☐ Brain Board ☒ IAR

☒ Board 12 ☐ CCE

SERVO_CENTER_PULSE_WIDTH .193

SERVO_RIGHT_PULSE_WIDTH .170

SERVO_LEFT_PULSE_WIDTH .216

ADC_SAMPLE_TIME 204.8

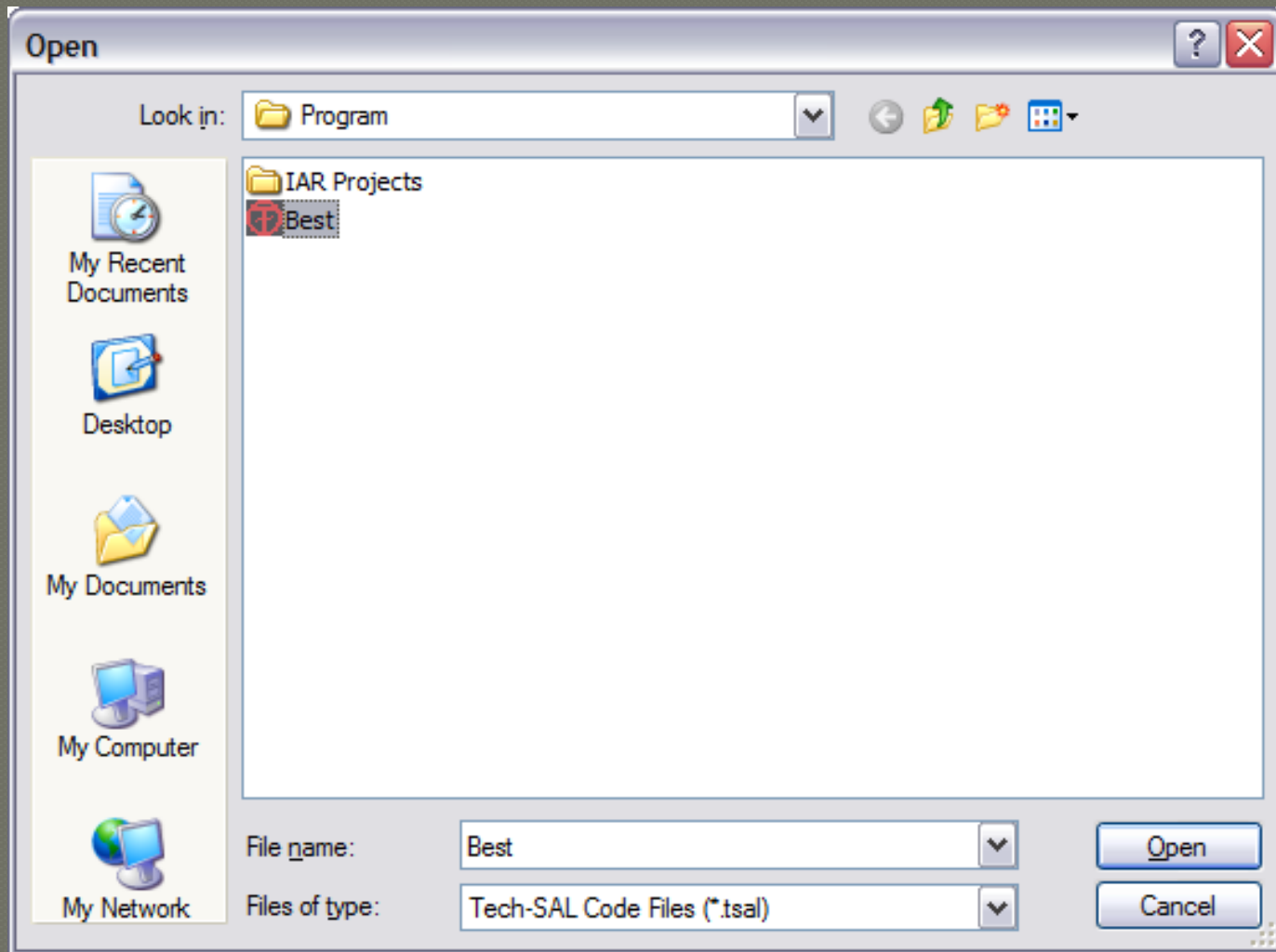
ADC_IR 0

ADC_IR 5

ADC_COLOR 3

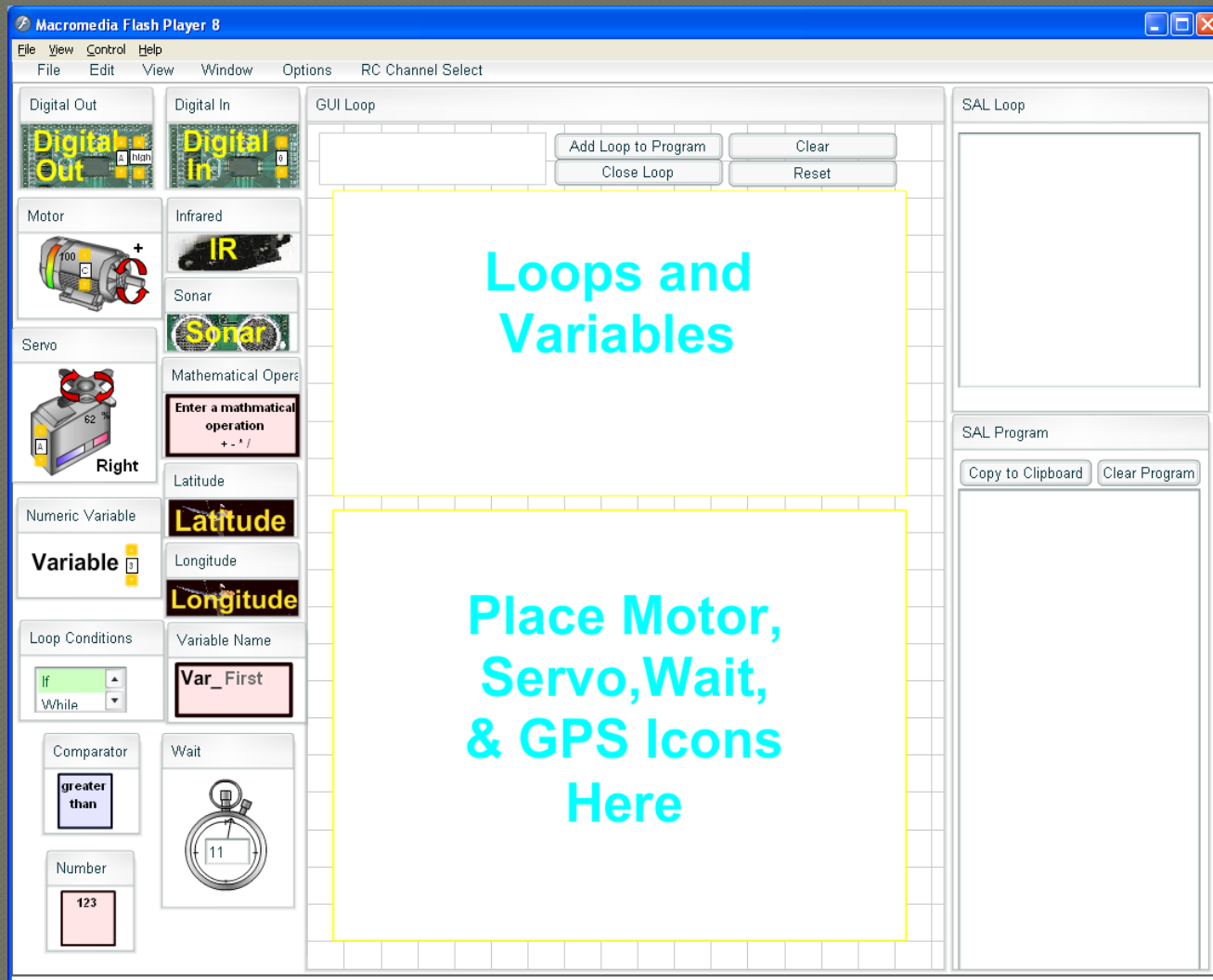
Generate SALConfig.txt Load SALConfig.txt Reset

Tech-SAL IDE: Open Dialog



Visual Tech-SAL Interface

Flash GUI v 2.0



Tech-Sal Versions

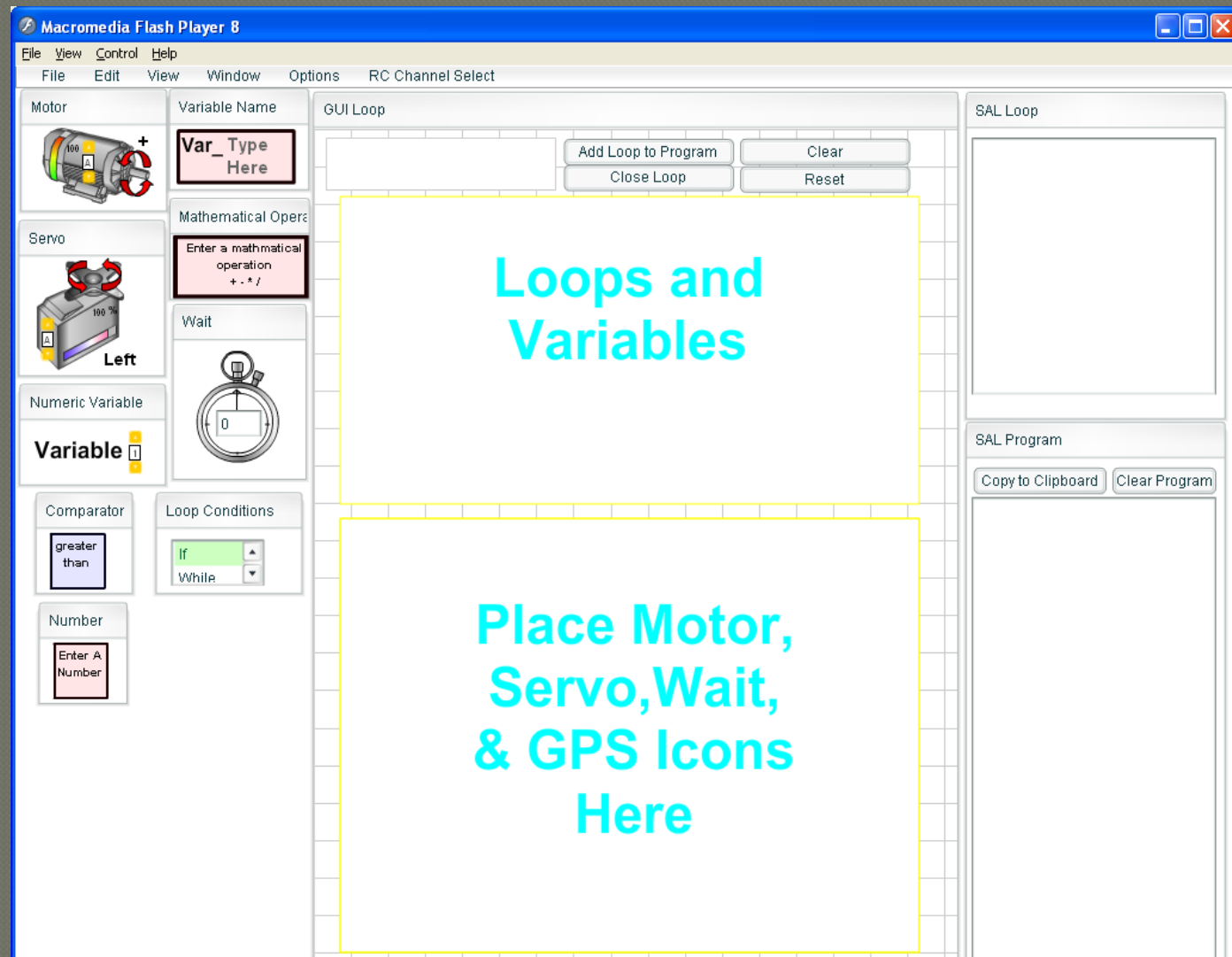
BEST Tech-Sal

- No Programming Experience Needed
- Only Basic Functions
 - No IR, Digital, GPS, Optical
- Brain Board Compatible

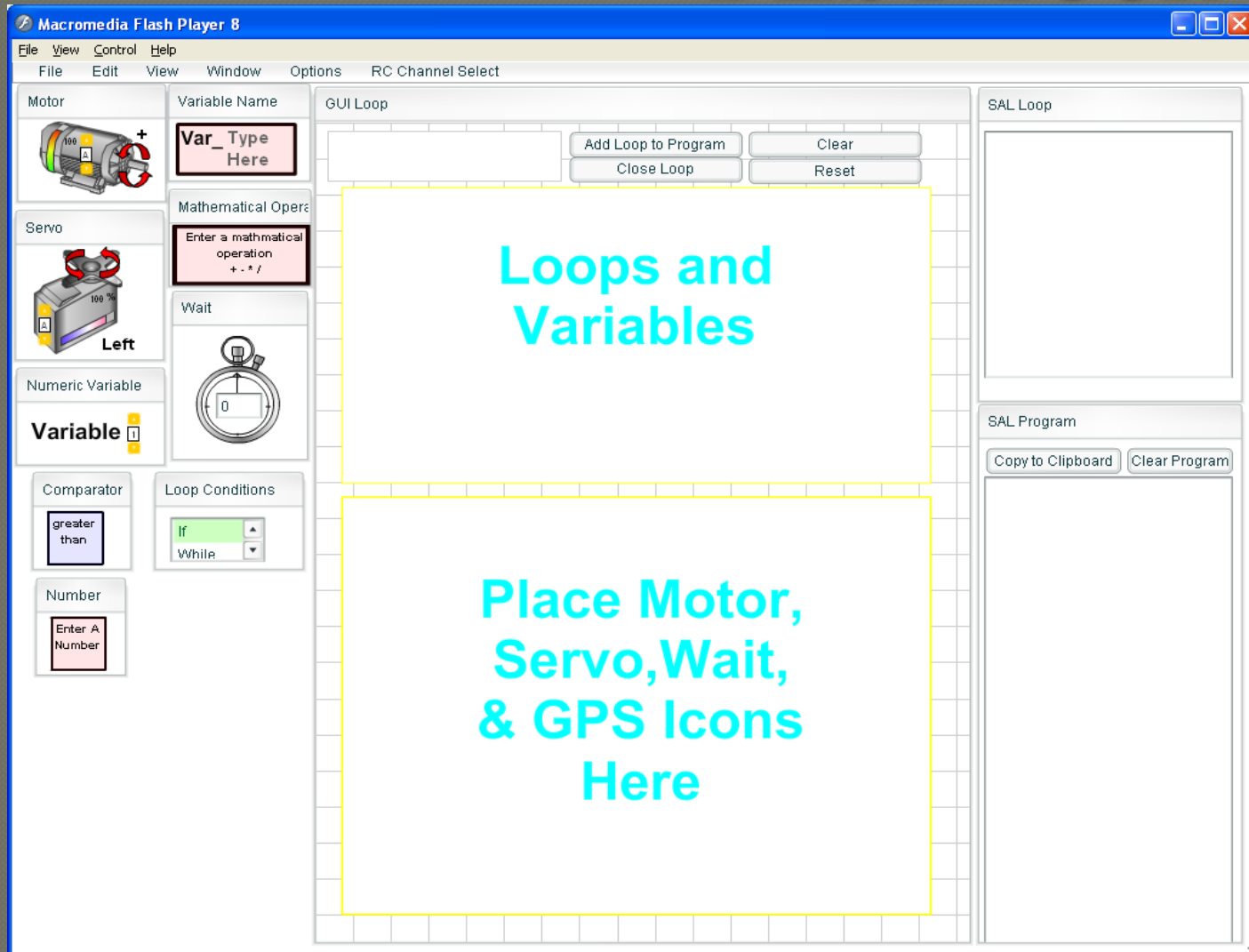
Full Tech-Sal

- Basic Programming Experience Needed
- Take Full Advantage of Hardware Functions
- Compatible with versions 8,10,12 and Brain Board

Full Tech-Sal



BEST Tech-Sal



GUI Improvements

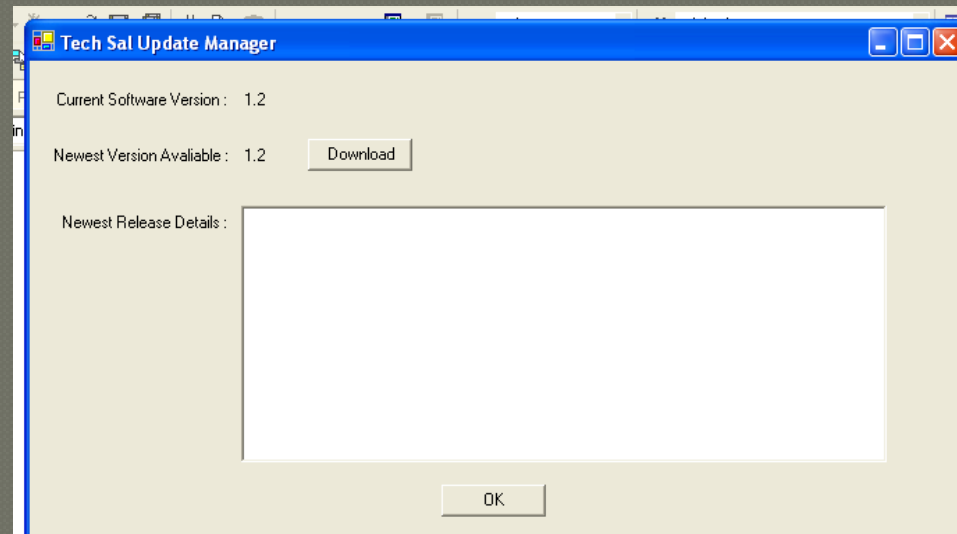
- Show Only Needed Icons In Best Version
- Create a Simple, Redistributable, and Installable Package
- Allow Addition of Non-Loop Code
- Change GUI Generated Code (. -> ;)
- Wrapped Up in Flash Projector (.exe)

Tech-Sal Update Tool

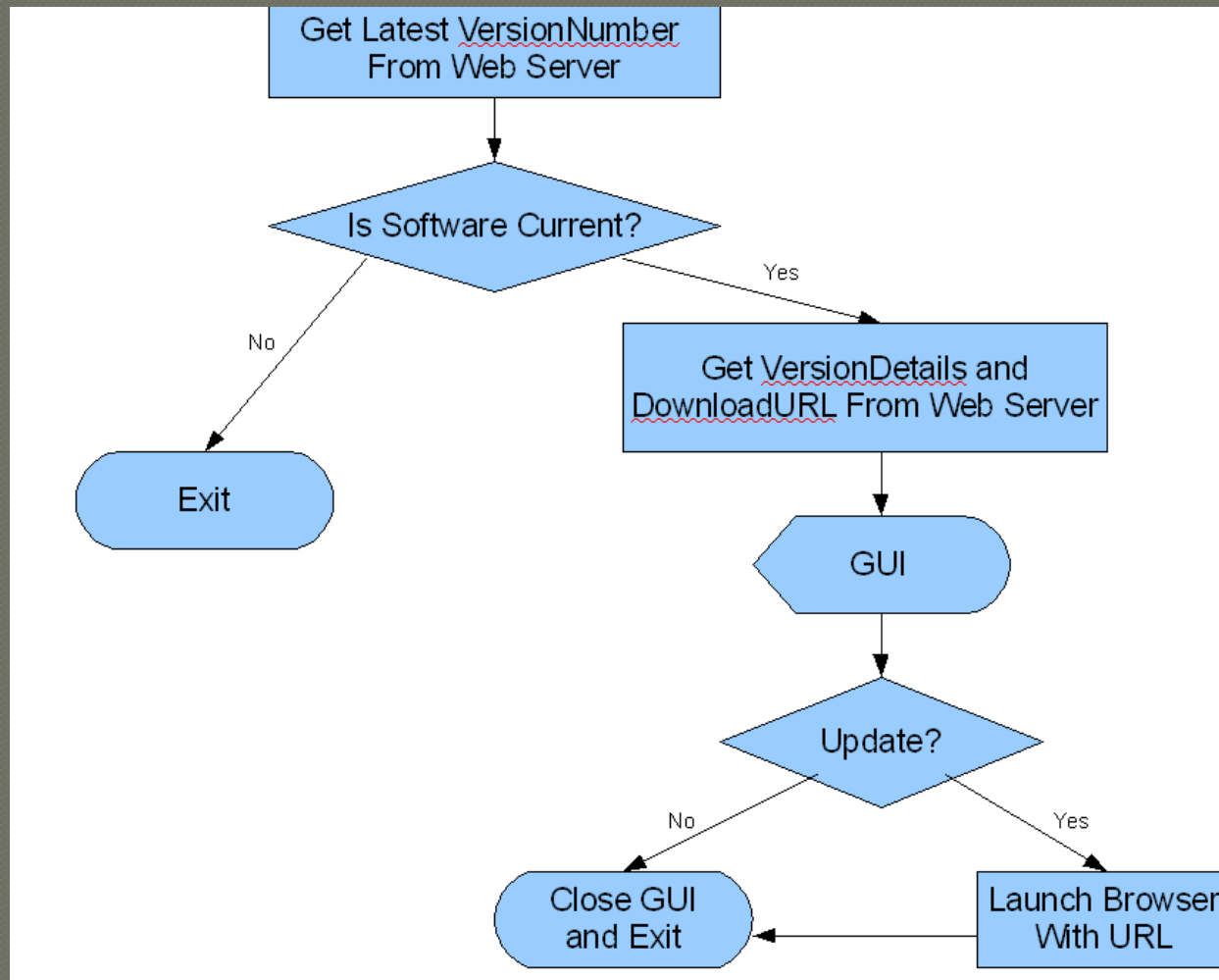
- Automate The Software Update Process
- BEST
- Integrate With IDE
- Check For Update Every Time IDE is Opened
- Non-Evasive if There is No update Available
- Alerts to New Version

Tech-Sal Update Tool

- Server/Client Based System
- Retrieve Newest Version (string)
- Retrieve Download URL and Release Details



Update Utility Flowchart (Client)



Documentation

- Visual Tech-SAL
 - Step-By-Step Instruction Manual
- Update Utility
 - Step by Step Server Setup Guide in IIS



CCE Compatability

IAR to CCE Migration

- ◉ Different intrinsics.h files.
- ◉ IAR has predefined interrupt state (istate_t)
- ◉ Different ISR structure.
- ◉ Registered defined differently.

Different intrinsics.h

```
#if defined(__IAR_SYSTEMS_ICC__)
    s=__get_interrupt_state();
    __disable_interrupt();
#elif defined(__TI_COMPILER_VERSION__)
    s=_get_SR_register();
    _disable_interrupts();
#endif
```

```
#if defined(__IAR_SYSTEMS_ICC__)
    __set_interrupt_state(s);
#elif defined(__TI_COMPILER_VERSION__)
    _bis_SR_register(s);
#endif
```

```
64  istate_t s;  
65  // protect the data structures  
66  s=__get_interrupt_state();
```

intrinsics.h from IAR

```
/*  
 * The return value of "__get_interrupt_state".  
 */  
  
typedef unsigned short istate_t;
```

Define istate_t for CCE

```
#if defined(__TI_COMPILER_VERSION__)  
    typedef unsigned short istate_t;  
#endif
```

Different ISR Structure

```
#if defined(__GNUC__)
interrupt(TIMERA1_VECTOR) Timer_A(void)
#elif defined(__IAR_SYSTEMS_ICC__)
#pragma vector=TIMERA1_VECTOR
__interrupt void Timer_A(void)
#elif defined(__TI_COMPILER_VERSION__)
__interrupt void Timer_A(void);
TIMERA1_ISR(Timer_A)
__interrupt void Timer_A(void)
#endif
```

IAR

CCE

In msp430x14x.h

```
#define TIMERA1_ISR(func)    ISR_VECTOR(func, ".int05")
```

Registered Defined Differently

```
#if defined(__IAR_SYSTEMS_ICC__)  
    volatile unsigned short *usCtlReg;  
#elif defined(__TI_COMPILER_VERSION__)  
    volatile unsigned int *usCtlReg;  
#endif
```

```
usCtlReg=&TBCCTL1;
```

Definition In Loop - CCE

Errors:

```
void user_timer(short time)
{
    for(short b=0; b<time;b++){
        for (long i=0;i<65300;i++) {
        }
    }
}
```

Compiles:

```
void user_timer(short time)
{
    short b;
    long i;
    for(b=0; b<time;b++){
        for (i=0;i<65300;i++) {
        }
    }
}
```

CCE Rejects Some Declarations

Error: Cannot declare variable before non-blocking assignment.

```
#if defined(__TI_COMPILER_VERSION__)
    int i;
#endif

void InitCpu(void)
{
    // Turn of watchdog for now implement
    WDTCTL=WDTPW+WDTHOLD;

    // Init the LED port early
    P1DIR |= BIT7;
    #if defined(__IAR_SYSTEMS_ICC__)
        int i;
    #endif
```

CCE vs. IAR – The Compilers

- CCE allows you to view all functions and variables in each file.
- IAR sets all variables to 0 when they are initialized, CCE does not.
- IAR has option to generate HEX code, for CCE you must use a command to call an external program.

IAR

Output file

☒ Override default

Secondary output file:

(None for the selected format)

BRAIN.txt

Format

☐ Debug information for C-SPY

☒ With runtime control modules

☒ With I/O emulation modules

☐ Buffered terminal output

☐ Allow C-SPY-specific extra output file

☒ Other

Output format: msp430-txt

Format variant: None

Module-local symbols: Include all

CCE

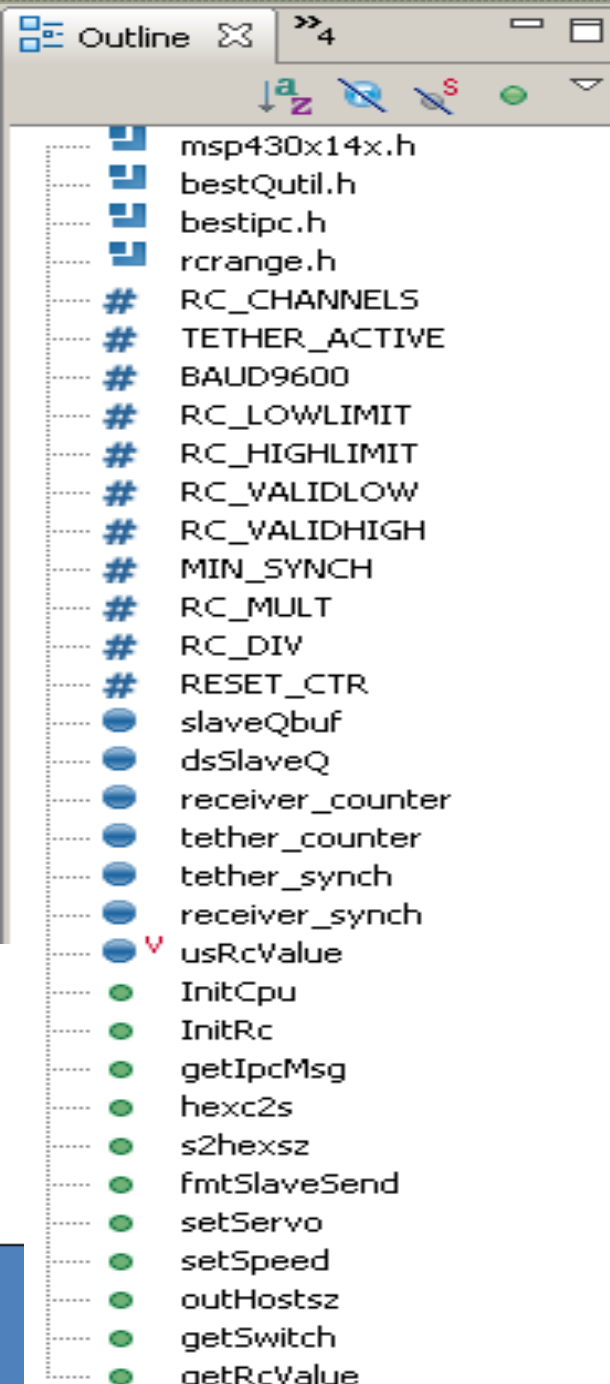
Post-build step:

Command:

`${CCE_INSTALL_ROOT}\tools\compiler\msp430\bin\hex430.exe --ti_txt brain.out -o brain.txt`

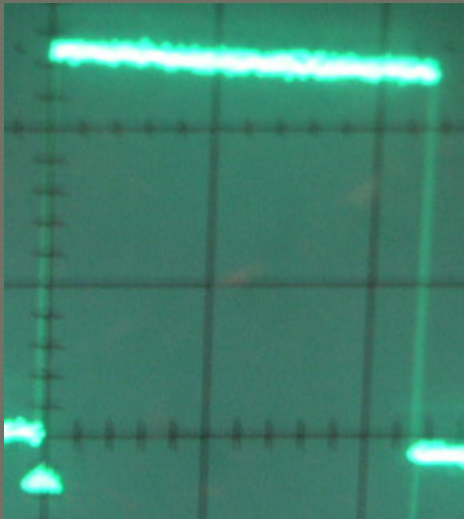
Description:

`${CCE_INSTALL_ROOT}\tools\compiler\msp430\bin\hex430.exe -ti_txt brain.out -o brain.txt`

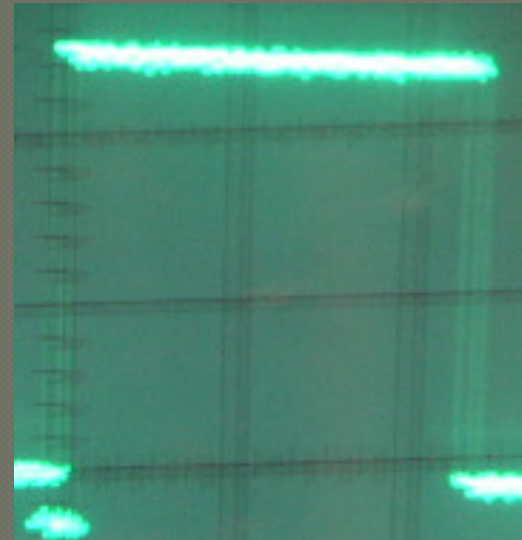


PWM Servo Test - Left

CCE



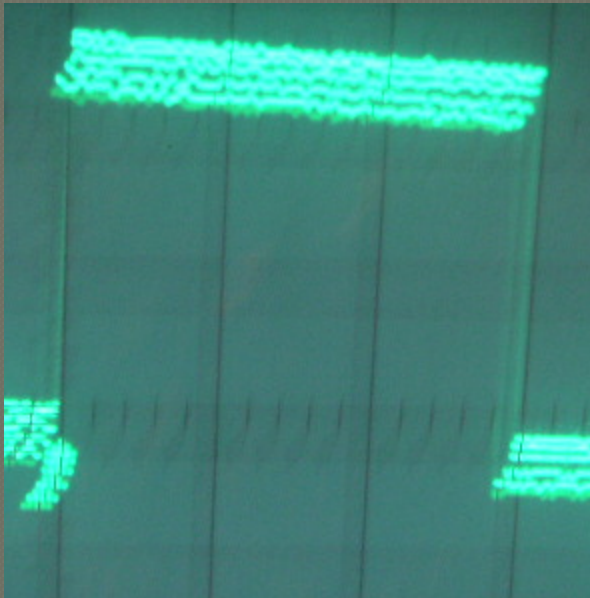
IAR



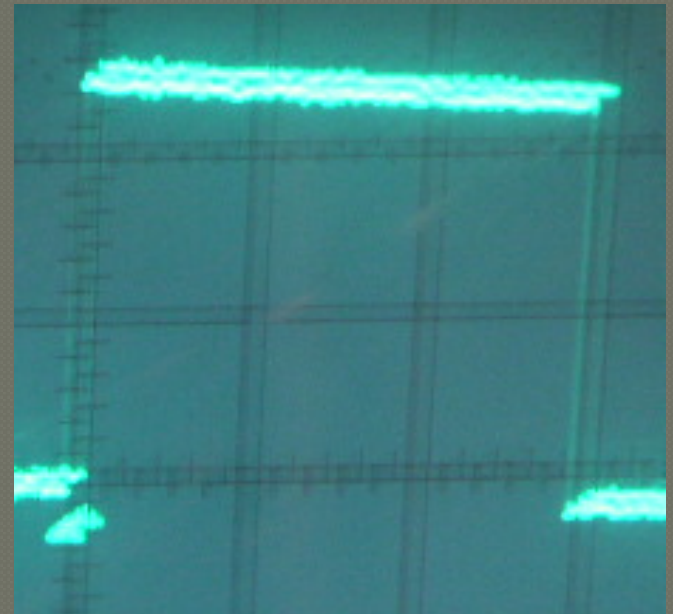
division - 500 μ s

PWM Servo Test - Center

CCE



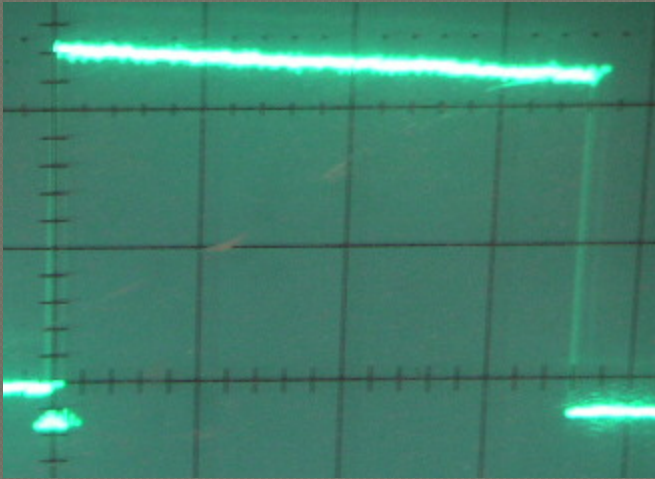
IAR



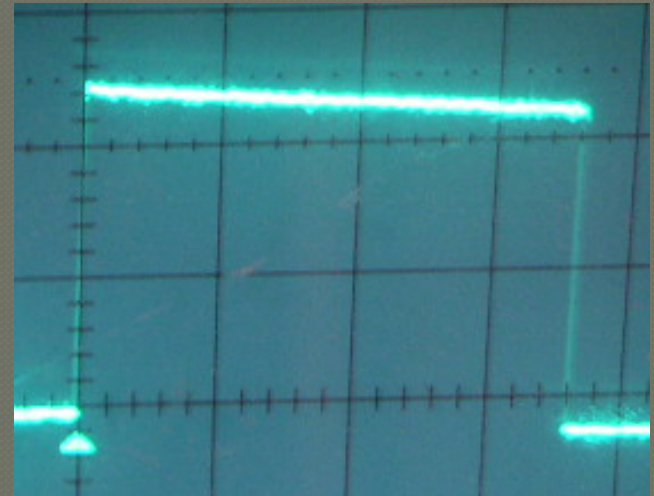
division - 500 μ s

PWM Servo Test - Right

CCE



IAR



division - 500 μ s

CCE vs. IAR - Preference

- ◉ Prefer the CCE User Interface
- ◉ Easier in IAR to generate TI-TXT files for loading programs.
- ◉ IAR is less picky about syntax.
- ◉ Debug mode runs smoother in IAR
- ◉ CCE generates smaller code than IAR.

Section I Accomplishments

Kyle's Deliverables

- ◉ Tech-SAL IDE
- ◉ Installation Package
- ◉ Quickstart Guide

Jacob's Deliverables

- ◉ Visual Tech-SAL GUI Improvements
- ◉ Update Tool
- ◉ Documentation Updating

Brian's Deliverables

- ◉ CCE Compatibility.
- ◉ One set of code that works in both IAR and CCE.
- ◉ Document migration for future developers.
- ◉ Document CCE vs. IAR for future boards.

Section II Accomplishments

Colby's Deliverables

This Semester

- ◉ Wrote code for multiple device mapping
- ◉ Coded the Syntax Checker from BNF rules
- ◉ Board 10/12 support
- ◉ BEST Presentation
- ◉ To-do list for next semester's group

Greg's Deliverables

◎ BOARD 12 USER FUNCTIONS:

- Board 12 Motor, Servo functions
- Board 12 H-Bridge ISR

◎ Tech-SAL Package

◎ BEST

- BEST and Tech-SAL
- BEST Electronic Scoring

Tim's Deliverables

- Code integration, maintenance and troubleshooting
- Coding/commenting rules
- Baukus-Naur syntax rules
- Syntax checker coding/testing
- Regional Texas BEST presentation
- Project Package for spring 2008

Budget / Gantt Chart

Original Gantt Chart

Gantt Chart

	Task Name	Start	Finish	Resource Name
1	SOFTWARE DEVELOPMENT	Tue 8/28/07	Mon 12/3/07	
2	WEEK 1	Tue 8/28/07	Mon 9/3/07	
3	Review Previous Final Reports	Tue 8/28/07	Mon 9/3/07	Kyle
4	Consult Previous Project Programmers	Tue 8/28/07	Mon 9/3/07	Colby
5	Review Current SAL Translator	Tue 8/28/07	Mon 9/3/07	Colby
6	Review Current SAL Syntax Detector	Tue 8/28/07	Mon 9/3/07	Tim
7	Review CCE Compatibility	Tue 8/28/07	Mon 9/3/07	Brian
8	Review Current Documentation	Tue 8/28/07	Mon 9/3/07	Greg
9	Review Current GUI	Tue 8/28/07	Mon 9/3/07	Jacob
10	WEEK 2	Tue 9/4/07	Mon 9/10/07	
11	Download and review Electronic Scoring Software	Tue 9/4/07	Mon 9/10/07	Tim
12	Brain Board RC Testing	Tue 9/4/07	Mon 9/10/07	Tim
13	Initial Testing-Back End (CCE Code)	Tue 9/4/07	Mon 9/10/07	Kyle
14	Initial Testing-Front End (GUI)	Tue 9/4/07	Mon 9/10/07	Greg
15	Revise CCE and RC Code	Tue 9/4/07	Mon 9/10/07	Brian
16	Revise SAL Translation Code	Tue 9/4/07	Mon 9/10/07	Colby
17	Revise GUI Code	Tue 9/4/07	Mon 9/10/07	Jacob
18	WEEK 3	Tue 9/11/07	Mon 9/17/07	
19	Brain Board RC Testing	Tue 9/11/07	Mon 9/17/07	Tim
20	Software Testing-Back End (CCE Code)	Tue 9/11/07	Mon 9/17/07	Kyle
21	Software Testing-Front End (GUI)	Tue 9/11/07	Mon 9/17/07	Greg
22	Revise GUI Code	Tue 9/11/07	Mon 9/17/07	Jacob
23	Revise SAL Translation Code	Tue 9/11/07	Mon 9/17/07	Colby
24	Revise CCE and RC Code	Tue 9/11/07	Mon 9/17/07	Brian
25	Compile First Revision	Tue 9/11/07	Mon 9/17/07	Tim

Gantt Chart

	Task Name	Start	Finish	Resource Names
26	WEEK 4	Tue 9/18/07	Mon 9/24/07	
27	Software Testing-Back End (CCE Code)	Tue 9/18/07	Mon 9/24/07	Kyle
28	Software Testing-Front End (GUI)	Tue 9/18/07	Mon 9/24/07	Greg
29	Software Documentation	Tue 9/18/07	Mon 9/24/07	Greg
30	Revise GUI Code	Tue 9/18/07	Mon 9/24/07	Jacob
31	Revise SAL Translation Code	Tue 9/18/07	Mon 9/24/07	Colby
32	Revise CCE Code	Tue 9/18/07	Mon 9/24/07	Brian
33	Compile and Upload Competition Version	Tue 9/18/07	Mon 9/24/07	Tim
34	WEEK 5	Tue 9/25/07	Mon 10/1/07	
35	Project Lab 2 Robot Testing	Tue 9/25/07	Mon 10/1/07	Kyle
36	Test Software	Tue 9/25/07	Mon 10/1/07	Greg
37	Revise GUI Code	Tue 9/25/07	Mon 10/1/07	Jacob
38	Revise SAL Translation Code	Tue 9/25/07	Mon 10/1/07	Colby
39	Revise CCE Code	Tue 9/25/07	Mon 10/1/07	Brian
40	Test Run of Electronic Scoring Software	Tue 9/25/07	Mon 10/1/07	Greg
41	SAL Software Support	Tue 9/25/07	Mon 10/1/07	Tim
42	WEEK 6	Tue 10/2/07	Mon 10/8/07	
43	Test Software with Lab 2 Robots	Tue 10/2/07	Mon 10/8/07	Kyle
44	Prepare Competition Scoring Software	Tue 10/2/07	Mon 10/8/07	Greg
45	Revise/Test GUI Code	Tue 10/2/07	Mon 10/8/07	Jacob
46	Acquire Scoring Setup (Projector, Laptop, etc)	Wed 10/3/07	Fri 10/5/07	Colby
47	Revise/Test SAL Translation Code	Tue 10/2/07	Mon 10/8/07	Colby
48	Revise/Test CCE Code	Tue 10/2/07	Mon 10/8/07	Brian
49	Compile and Upload New Version	Tue 10/2/07	Mon 10/8/07	Tim

	Task Name	Start	Finish	Resource Name
50	WEEK 7	Tue 10/9/07	Mon 10/15/07	
51	Test Software with Lab 2 Robots	Tue 10/9/07	Mon 10/15/07	Kyle
52	Upload Software Revisions	Tue 10/9/07	Mon 10/15/07	Greg
53	Revise/Test GUI Code	Tue 10/9/07	Mon 10/15/07	Jacob
54	Revise/Test SAL Translation Code	Tue 10/9/07	Mon 10/15/07	Colby
55	Revise/Test CCE Code	Tue 10/9/07	Mon 10/15/07	Brian
56	SAL Software Support	Tue 10/9/07	Mon 10/15/07	Tim
57	Prepare Interim Presentation	Tue 10/9/07	Mon 10/15/07	Team
58	Prepare Interim Reports	Tue 10/9/07	Mon 10/15/07	Individual
59	WEEK 8	Tue 10/16/07	Mon 10/22/07	
60	Test Revisions with Lab 2 Robots	Tue 10/16/07	Mon 10/22/07	Kyle
61	SAL Software Support	Tue 10/16/07	Mon 10/22/07	Greg
62	Revise/Test GUI Code	Tue 10/16/07	Mon 10/22/07	Jacob
63	Acquire Scoring Setup (Project, Laptop, etc)	Thu 10/18/07	Fri 10/19/07	Colby
64	Revise/Test SAL Translation Code	Tue 10/16/07	Mon 10/22/07	Colby
65	Revise/Test CCE Code	Tue 10/16/07	Mon 10/22/07	Brian
66	Compile and Upload New Version	Tue 10/16/07	Mon 10/22/07	Tim
67	WEEK 9	Tue 10/23/07	Mon 10/29/07	
68	Revise Documentation	Tue 10/23/07	Mon 10/29/07	Kyle
69	Test Software	Tue 10/23/07	Mon 10/29/07	Greg
70	Revise GUI Code	Tue 10/23/07	Mon 10/29/07	Jacob
71	Revise SAL Translation Code	Tue 10/23/07	Mon 10/29/07	Colby
72	Revise CCE Code	Tue 10/23/07	Mon 10/29/07	Brian
73	SAL Software Support	Tue 10/23/07	Mon 10/29/07	Tim

	Task Name	Start	Finish	Resource Names
74	WEEK 10	Tue 10/30/07	Mon 11/5/07	
75	SAL Software Support	Tue 10/30/07	Mon 11/5/07	Kyle
76	Revise Documentation	Tue 10/30/07	Mon 11/5/07	Greg
77	Test GUI Code	Tue 10/30/07	Mon 11/5/07	Jacob
78	Test SAL Translation Code	Tue 10/30/07	Mon 11/5/07	Colby
79	Test CCE Code	Tue 10/30/07	Mon 11/5/07	Brian
80	Compile and Upload New Version	Tue 10/30/07	Mon 11/5/07	Tim
81	WEEK 11	Tue 11/6/07	Mon 11/12/07	
82	Revise Documentation	Tue 11/6/07	Mon 11/12/07	Kyle
83	Test Software	Tue 11/6/07	Mon 11/12/07	Greg
84	Revise GUI Code	Tue 11/6/07	Mon 11/12/07	Jacob
85	Revise SAL Translation Code	Tue 11/6/07	Mon 11/12/07	Colby
86	Revise CCE Code	Tue 11/6/07	Mon 11/12/07	Brian
87	SAL Software Support	Tue 11/6/07	Mon 11/12/07	Tim
88	WEEK 12	Tue 11/13/07	Mon 11/19/07	
89	SAL Software Support	Tue 11/13/07	Mon 11/19/07	Kyle
90	Revise Documentation	Tue 11/13/07	Mon 11/19/07	Greg
91	Test GUI Code	Tue 11/13/07	Mon 11/19/07	Jacob
92	Test SAL Translation Code	Tue 11/13/07	Mon 11/19/07	Colby
93	Test CCE Code	Tue 11/13/07	Mon 11/19/07	Brian
94	Compile and Upload New Version	Tue 11/13/07	Mon 11/19/07	Tim

95	WEEK 13	Tue 11/20/07	Mon 11/26/07	
96	SAL Software Support	Tue 11/20/07	Mon 11/26/07	Kyle
97	Finalize Documentation	Tue 11/20/07	Mon 11/26/07	Greg
98	Finalize GUI Code	Tue 11/20/07	Mon 11/26/07	Jacob
99	Finalize SAL Translation Code	Tue 11/20/07	Mon 11/26/07	Colby
100	Finalize CCE Code	Tue 11/20/07	Mon 11/26/07	Brian
101	Compile and Upload Final Version	Tue 11/20/07	Mon 11/26/07	Tim
102	WEEK 14	Tue 11/27/07	Mon 12/3/07	
103	Final Report	Tue 11/27/07	Mon 12/3/07	Team
104	Final Presentation	Tue 11/27/07	Mon 12/3/07	Team
105	Project Notebook	Tue 11/27/07	Mon 12/3/07	Individual

Final Gantt Chart

WEEK 1		Name : SOFTWARE DEVELOPMENT		8/28/07	9/4/07	WEEK 3		9/11/07	9/18/07
Review Previous Final Reports		8/28/07	9/4/07			Tested RC Code from SAL and BEST		9/11/07	9/18/07
Consult Previous Project Programmers		8/28/07	9/4/07			Create GUI User Documentation		9/11/07	9/18/07
Review Current SAL Translator		8/28/07	9/4/07			Review Schematics for Board 12 Moto...		9/11/07	9/18/07
Review Current SAL Syntax Detector		8/28/07	9/4/07			Create BEST GUI Version		9/11/07	9/18/07
Review CCE Compatibility		8/28/07	9/4/07			Begin modifying SAL to translate RC C...		9/11/07	9/18/07
Review Current Documentation		8/28/07	9/4/07			Revise CCE and RC Code		9/11/07	9/18/07
Review Current GUI		8/28/07	9/4/07			WEEK 4		9/18/07	9/25/07
WEEK 2		9/4/07	9/11/07			Create SAL Package		9/18/07	9/25/07
Electronic Scoring Setup (Hardware)		9/4/07	9/11/07			Create Board 12 Motor and Servo Code		9/18/07	9/25/07
Electronic Scoring Setup (Software)		9/4/07	9/11/07			Revise SAL Tranlation for BRAIN board		9/18/07	9/25/07
Fix BEST Demo Robot for Kick-Off		9/4/07	9/11/07			Create GUI Versions (BEST, SAL)		9/18/07	9/25/07
Revise CCE and RC Code		9/4/07	9/11/07			Test Motor/Servo Outputs for Brain B...		9/18/07	9/25/07
Revise SAL Translation Code		9/4/07	9/11/07			Revise CCE Code		9/18/07	9/25/07
Revise GUI Code		9/4/07	9/11/07						

WEEK 5	9/25/07	10/2/07	WEEK 8	10/16/07	10/23/07
.....Add Control Set to GUI (Brain, Board 12)	9/25/07	10/2/07	Revise SAL Package	10/16/07	10/23/07
.....ReWrite Motor (B12), header files	9/25/07	10/2/07	SAL Software Support	10/16/07	10/23/07
.....Change 'period' to 'semi-colon' for SAL...	9/25/07	10/2/07	Revise/Test GUI Code	10/16/07	10/23/07
.....Differential Steering Mapping w/ Brain ...	9/25/07	10/2/07	Acquire Scoring Setup (Project, Lapt...	10/18/07	10/20/07
.....Finish CCE Code	9/25/07	10/2/07	Revise/Test SAL Translation Code	10/16/07	10/23/07
.....Modify 'for' loop statement code	9/25/07	10/2/07	Revise/Test CCE Code	10/16/07	10/23/07
WEEK 6	10/2/07	10/9/07	Compile and Upload New Version	10/16/07	10/23/07
.....Finalize SAL Package	10/2/07	10/9/07	WEEK 9	10/23/07	10/30/07
.....Finish Board Header Files	10/2/07	10/9/07	Revise Documentation	10/23/07	10/30/07
.....Add Save/Open Function to GUI	10/2/07	10/9/07	Test Software	10/23/07	10/30/07
.....Acquire Scoring Setup (Projector, Lapt...	10/3/07	10/6/07	Revise GUI Code	10/23/07	10/30/07
.....Continue Differential Steering Mappin...	10/2/07	10/9/07	Revise SAL Translation Code	10/23/07	10/30/07
.....Board Testing	10/2/07	10/9/07	Revise CCE Code	10/23/07	10/30/07
.....Draft SAL commenting and code conv...	10/2/07	10/9/07	SAL Software Support	10/23/07	10/30/07
WEEK 7	10/9/07	10/16/07	WEEK 10	10/30/07	11/6/07
.....Complete SAL IDE	10/9/07	10/16/07	Tech-SAL Installation Package	10/30/07	11/6/07
.....Board 12 H-Bridge Interrupt	10/9/07	10/16/07	Create BEST Header File	10/30/07	11/6/07
.....Revise/Test GUI Code	10/9/07	10/16/07	Work on Update Manager	10/30/07	11/6/07
.....Finish BRAIN RC Differential Mapping	10/9/07	10/16/07	SAL Syntax Checker	10/30/07	11/6/07
.....Revise/Test CCE Code	10/9/07	10/16/07	Test CCE Code	10/30/07	11/6/07
.....Continue SAL commenting and code re...	10/9/07	10/16/07	SAL Syntax Rules	10/30/07	11/6/07
.....Prepare Interim Presentation	10/9/07	10/16/07			
.....Prepare Interim Reports	10/9/07	10/16/07			

[-] WEEK 12	11/13/07	11/20/07
Compile Tech-SAL Package for Regionals	11/13/07	11/20/07
Test Autonomous and RC Brain Functi...	11/13/07	11/20/07
Finalize Update Manager	11/13/07	11/20/07
SAL Syntax Checker	11/13/07	11/20/07
Finalize CCE/IAR Compatibility Issues	11/13/07	11/20/07
Finish SAL Syntax Checker Helper Fun...	11/13/07	11/20/07
[-] WEEK 13	11/20/07	11/27/07
Compile Tech-SAL Package for Regionals	11/20/07	11/27/07
Finalize Translation Code	11/20/07	11/27/07
Finalize GUI	11/20/07	11/27/07
SAL Syntax Checker	11/20/07	11/27/07
Add CCE/IAR Compatibility	11/20/07	11/27/07
Finalize Tech-SAL Package	11/20/07	11/27/07
[-] WEEK 14	11/27/07	12/4/07
Final Report	11/27/07	12/4/07
Final Presentation	11/27/07	12/4/07
Project Notebook	11/27/07	12/4/07

Questions?